

# Conversion Practice

|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Volume & Capacity | 2 liters = _____ milliliters<br>_____ = _____                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 3 quarts = <u>6</u> pints<br>$\begin{array}{r} \downarrow \div 3 \\ 3 \text{ qt} \\ \times \text{ pt} \end{array} = \begin{array}{r} \downarrow \times 2 \\ 1 \text{ qt} \\ 2 \text{ pt} \end{array}$<br>SF: $\times 2$                                                                                                                                                                                                                                                                                                                                                                                    |
| Length            | 75 centimeters = <u>0.75</u> meters<br>$\begin{array}{r} \downarrow \div 100 \\ 75 \text{ cm} \\ \times \text{ m} \end{array} = \begin{array}{r} \downarrow \div 100 \\ 100 \text{ cm} \\ 1 \text{ m} \end{array}$<br>SF: $\times \frac{1}{100}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 12 feet = <u>4</u> yards<br>$\begin{array}{r} \downarrow \div 3 \\ 12 \text{ ft} \\ \times \text{ yd} \end{array} = \begin{array}{r} \downarrow \div 3 \\ 3 \text{ ft} \\ 1 \text{ yd} \end{array}$<br>SF: $\times \frac{1}{3}$                                                                                                                                                                                                                                                                                                                                                                            |
| Weight & Mass     | 6 pounds = <u>96</u> ounces<br>$\begin{array}{r} \downarrow \times 16 \\ 6 \text{ lb} \\ \times \text{ oz} \end{array} = \begin{array}{r} \downarrow \times 16 \\ 1 \text{ lb} \\ 16 \text{ oz} \end{array}$<br>SF: $\times 16$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2,500 grams = <u>2.5</u> kilograms<br>$\begin{array}{r} \downarrow \div 1000 \\ 2,500 \text{ g} \\ \times \text{ kg} \end{array} = \begin{array}{r} \downarrow \div 1000 \\ 1,000 \text{ g} \\ 1 \text{ kg} \end{array}$<br>SF: $\times \frac{1}{1000}$                                                                                                                                                                                                                                                                                                                                                    |
| Snake Method      | <p>What if the STAAR chart does not give you a direct conversion? Use the "snake method"</p> <p>3 gallons = <math>3 \times 16 = 48</math> cups</p> <p>SF: <math>4 \times 2 \times 2 = \textcircled{\times 16}</math></p> <p>Customary</p> <p>1 gallon (gal) = 4 quarts (qt)</p> <p>1 quart (qt) = 2 pints (pt)</p> <p>1 pint (pt) = 2 cups (c)</p> $\begin{array}{r} \downarrow \times 4 \\ 3 \text{ gal} \\ \times \end{array} = \begin{array}{r} \downarrow \times 4 \\ 1 \text{ gal} \\ 4 \text{ qt} \end{array}$ $\begin{array}{r} \downarrow \times 2 \\ 12 \text{ qt} \\ \times \end{array} = \begin{array}{r} \downarrow \times 2 \\ 1 \text{ qt} \\ 2 \text{ pt} \end{array}$ $\begin{array}{r} \downarrow \times 2 \\ 24 \text{ pt} \\ \times \end{array} = \begin{array}{r} \downarrow \times 2 \\ 1 \text{ pt} \\ 2 \text{ cup} \end{array}$ | <p>Snake Method:</p> <p>2 miles = <u>10,560</u> feet</p> <p>SF: <math>\times 1760 \times 3 = \textcircled{\times 5280}</math></p> <p>Customary</p> <p>1 mile (mi) = 1,760 yards (yd)</p> <p>1 yard (yd) = 3 feet (ft)</p> <p>1 foot (ft) = 12 inches (in.)</p> $\begin{array}{r} \downarrow \times 1760 \\ 2 \text{ mi} \\ \times \end{array} = \begin{array}{r} \downarrow \times 1760 \\ 1 \text{ mi} \\ 1760 \text{ yd} \end{array}$ $\begin{array}{r} \downarrow \times 3 \\ 3520 \text{ yd} \\ \times \end{array} = \begin{array}{r} \downarrow \times 3 \\ 1 \text{ yd} \\ 3 \text{ ft} \end{array}$ |

**Problem-Solving Practice: Convert Measurement Units**

1. The average weight of a baby at birth is 7 pounds. How many ounces is the average weight of a baby?

$$\begin{array}{r} 7 \text{ lb} \\ \times 16 \text{ oz} \\ \hline \end{array} = \begin{array}{r} 1 \text{ lb} \\ \times 16 \text{ oz} \\ \hline \end{array}$$

Handwritten work shows a conversion factor of 16 oz per 1 lb, and a calculation of 7 lb multiplied by 16 to get 112 oz.

$$X = 112 \text{ ounces}$$

2. The height of Niagara Falls is 56 meters. How many kilometers high is Niagara Falls?

$$\begin{array}{r} 56 \text{ m} \\ \div 1,000 \\ \hline \end{array} = \begin{array}{r} 1,000 \text{ m} \\ \div 1,000 \\ \hline \end{array}$$

Handwritten work shows a conversion factor of 1,000 m per 1 km, and a calculation of 56 m divided by 1,000 to get 0.056 km.

$$X = 0.056 \text{ km}$$

3. The gasoline tank of a minivan holds 18 gallons. How many quarts can the tank hold?

$$\begin{array}{r} 18 \text{ gal} \\ \times 4 \text{ qt} \\ \hline \end{array} = \begin{array}{r} 1 \text{ gal} \\ \times 4 \text{ qt} \\ \hline \end{array}$$

Handwritten work shows a conversion factor of 4 qt per 1 gal, and a calculation of 18 gal multiplied by 4 to get 72 qt.

$$X = 72 \text{ quarts}$$

4. Cell phones can weigh as little as 56 grams. How many kilograms can the cell phone weigh?

$$\begin{array}{r} 56 \text{ g} \\ \div 1,000 \\ \hline \end{array} = \begin{array}{r} 1,000 \text{ g} \\ \div 1,000 \\ \hline \end{array}$$

Handwritten work shows a conversion factor of 1,000 g per 1 kg, and a calculation of 56 g divided by 1,000 to get 0.056 kg.

$$X = 0.056 \text{ kg}$$

5. A recipe for ice cream calls for 1.656 liters of milk. How many milliliters of milk are in the recipe?

$$\begin{array}{r} 1.656 \text{ L} \\ \times 1,000 \\ \hline \end{array} = \begin{array}{r} 1 \text{ L} \\ \times 1,000 \\ \hline \end{array}$$

Handwritten work shows a conversion factor of 1,000 mL per 1 L, and a calculation of 1.656 L multiplied by 1,000 to get 1,656 mL.

$$X = 1,656 \text{ mL}$$

6. The Statue of Liberty weighs 450,000 pounds. How many tons does the statue weigh?

$$\begin{array}{r} 450,000 \text{ lb} \\ \div 2,000 \\ \hline \end{array} = \begin{array}{r} 2,000 \text{ lb} \\ \div 2,000 \\ \hline \end{array}$$

Handwritten work shows a conversion factor of 2,000 lb per 1 ton, and a calculation of 450,000 lb divided by 2,000 to get 225 tons.

$$X = 225 \text{ tons}$$

7. The Ted Williams Tunnel under Boston Harbor is 2.57 kilometers long. How many meters is the length of the tunnel?

$$\begin{array}{r} 2.57 \text{ km} \\ \times 1,000 \\ \hline \end{array} = \begin{array}{r} 1 \text{ km} \\ \times 1,000 \\ \hline \end{array}$$

Handwritten work shows a conversion factor of 1,000 m per 1 km, and a calculation of 2.57 km multiplied by 1,000 to get 2,570 m.

$$X = 2,570 \text{ m}$$

8. The United States exports over 200 billion pounds of coal. How many tons does the United States export?

$$\begin{array}{r} 200,000,000,000 \text{ lb} \\ \div 2,000 \\ \hline \end{array} = \begin{array}{r} 2,000 \text{ lb} \\ \div 2,000 \\ \hline \end{array}$$

Handwritten work shows a conversion factor of 2,000 lb per 1 ton, and a calculation of 200,000,000,000 lb divided by 2,000 to get 100,000,000 tons.

$$X = 100,000,000 \text{ tons}$$